



Solution Set

Question 1

A unique solution exists if the coefficient matrix is invertible. Its determinant is

$$\det \begin{bmatrix} 1 & 2 & -3 \\ 2 & 6 & -11 \\ 1 & -2 & 7 \end{bmatrix} = 0$$

so the system never has a unique solution for any a, b, c. Depending on a, b, c it has either no solution or infinitely many solutions.

Question 2

For

$$A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 0 & 1 \\ 3 & 2 & 1 \end{bmatrix}$$

the minor matrix is

$$M = \begin{bmatrix} -2 & -4 & -2 \\ 4 & 4 & -4 \\ 2 & 0 & 2 \end{bmatrix}$$

The cofactor matrix is

$$C = \begin{bmatrix} -2 & 4 & -2 \\ -4 & 4 & 4 \\ 2 & 0 & 2 \end{bmatrix}$$

Question 3

Because the third row has only the last entry nonzero,

$$\begin{aligned} |A| &= (t-4) \det \begin{bmatrix} t-2 & 4 \\ 1 & t+1 \end{bmatrix} \\ |A| &= (t-4) \{ (t-2)(t+1) - 4 \} = (t-4)(t^2 - t - 6) \\ |A| &= (t-4)(t-3)(t+2) \end{aligned}$$

Thus

$$t \in \{-2, 3, 4\}$$

Question 4

Projection onto the line spanned by

$$a = (2, 1, 3)^\top$$

is

$$P = \frac{aa^\top}{a^\top a} = \frac{1}{14} \begin{bmatrix} 4 & 2 & 6 \\ 2 & 1 & 3 \\ 6 & 3 & 9 \end{bmatrix}$$

Column space:

$$\text{Col}(P) = \text{span} \{ (2, 1, 3)^\top \}$$

Nullspace:

$$\text{Null}(P) = \{ x : 2x_1 + x_2 + 3x_3 = 0 \}$$

one basis is

$$\{ (-1, 2, 0)^\top, (-3, 0, 2)^\top \}$$

Eigenvalue 1 is along the projection line; eigenvalue 0 is for every vector in the nullspace.

Question 5

A 3 by 5 matrix with rank 3 has full row rank, so its column space is

$$\mathbb{R}^3$$

Therefore

$$Ax = b$$

always has a solution for every

$$b \in \mathbb{R}^3$$

but it cannot be unique because there are 5 unknowns and rank 3. The answer is: always has many solutions. The nullspace has dimension

$$5 - 3 = 2$$